Introduction to Statistics

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Agenda

- Types of Data
  - Qualitative vs quantitative data
  - Understanding the variables
    - Numerical Variable
    - Categorical Variable

- Types of statistics
- Parameter vs statistics

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Agenda

- Sampling Techniques
  - Probabilistic Sampling
    - Simple Random Sampling (with and without replacement)
  - Resampling
    - Bootstrap resampling
    - Jackknife resampling

- Descriptive Statistics and visualization
  - Measure of Central Tendency
  - Distribution of the data
  - Measure of Dispersion
  - Skewness and Kurtosis
  - Covariance and Correlation

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The Data

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What is Data?

Data is units of information in structured or unstructured format.

Examples:

- Collection of relevant tweets
- Records of yield in a farm over a period of time
- Records of stock price every minute
- Records of performance of a sports person

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Types of Data

In general, there are four types of data. They are:

- Qualitative data vs Quantitative data
- Structured data vs unstructured data

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Understanding the variable

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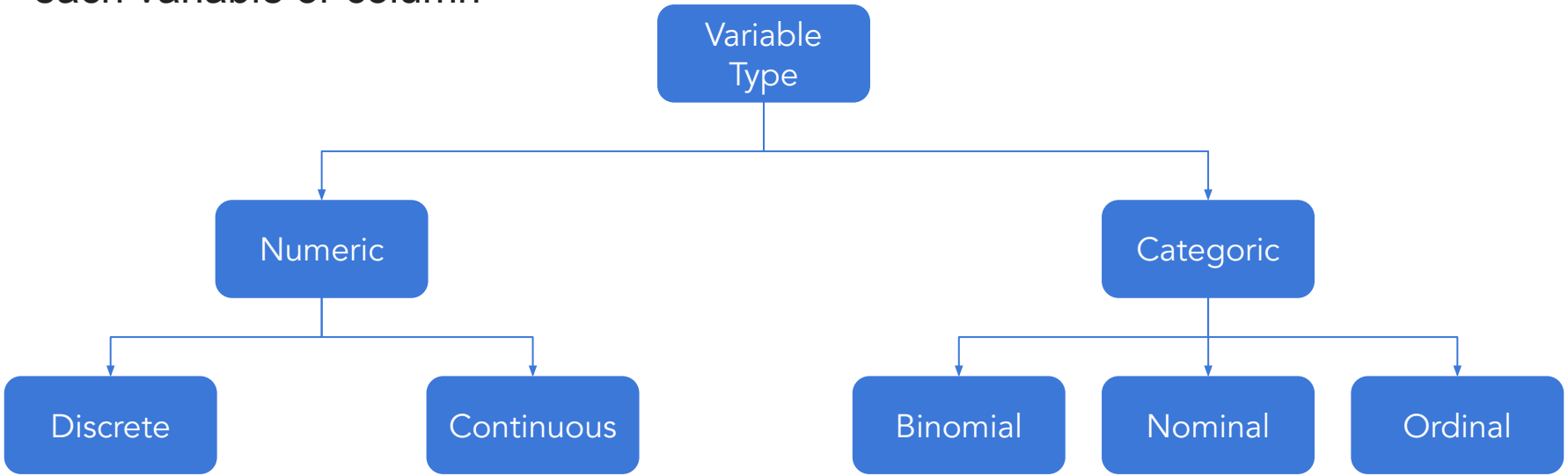
Understand the variable

To understand the descriptive statistics, we need to first understand the type of each variable or column



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Numerical Variable

- A variable which takes a numeric value is called a numeric variable
- Also known as quantitative variable
- A discrete numeric variable is a random variable which takes discrete values, i.e. values from the set of whole numbers only. It can take countably finite values

Examples:

1. Number of cars passing by a toll-gate every one minute
2. Number of defective items in a box

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Numerical Variable

A continuous numeric variable is a variable which can have infinite number of values within a range

Examples:

1. The amount of rainfall in millimeters
2. Price of a stock

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Categorical Variable

- Categorical variable has two or more levels
- Also known as qualitative variable

Examples:

1. Colors: red, orange, yellow
2. Gender: male, female
3. Economic status: low, medium, high

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Categorical Variable

Nominal Data:

- Nominal data has no order and has two or more than two categories
- It represents discrete units and are used to label variables
- Example: Housing type - Apartment, Bungalow, Penthouse

Binary Data:

- Binary data has no order and has strictly two categories
- Also known as dichotomous variable
- Example: Presence or absence of a disease

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Categorical Variable

Ordinal Data:

- Ordinal data is ordered nominal data
- It represents discrete and ordered units
- Example: Education - Primary, Secondary, High School, College and University which are ordered

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PLEASE!! NOTE

- At times data may appear to be numeric but is actually categorical

- Consider the adjoining table. The ‘Gender’ column has value 1 and 0

- The numeric values used to represent a categorical variable cannot be used for numerical computation

Name	Gender	Age
Ria	1	45
Arya	1	23
Sam	0	34
Joe	0	54
Harry	0	12
Sarah	1	43

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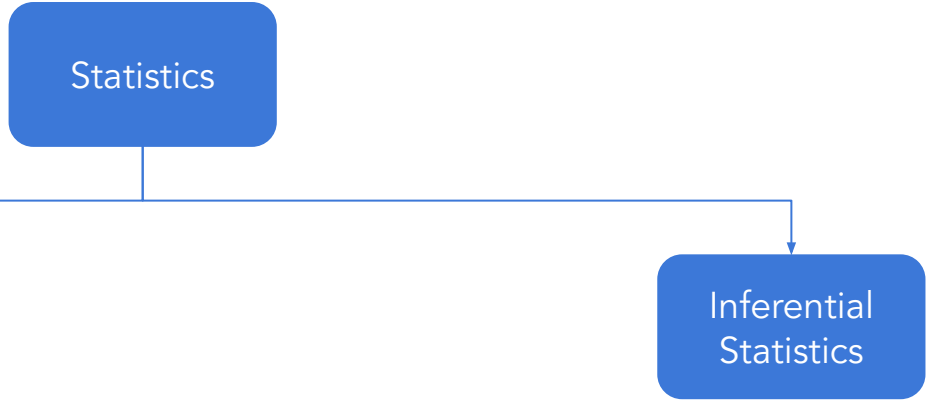
Types of Statistics



It is used to only describe the sample or summarize information about the sample

It is used to make inferences and generalizations about the broader population

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Parameter and Statistic

- Parameter:
  - The population quantities are termed as a parameter
  - In inferential statistics, sample is used to estimate the population parameters
  - Example: Population mean

- Statistic:
  - Statistical measure computed from sample observations
  - Also known as an estimator
  - Example: Sample mean

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Parameter and Statistic

The statistical notations for population parameters and sample statistics

	Population Parameter	Sample Statistic
Mean	μ	$\bar{x}$
Variance	σ^2	s^2
Standard Deviation	σ	s
Proportion	p	$\bar{p}$

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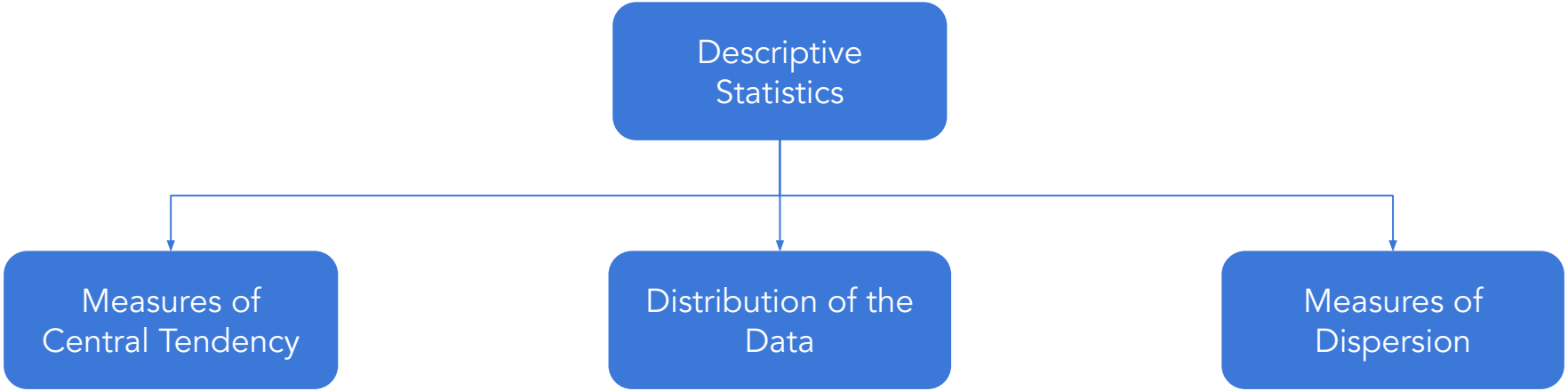
Descriptive Statistics

Descriptive statistics is the term given to the analysis of the data that helps describe, visualize or summarize the data



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Exploratory Data Analysis (EDA)

- EDA is a process of analyzing the datasets to summarize their main features using numerical and visual methods

- In this session, we shall cover the numerical methods to analyze the data

- The numerical methods includes descriptive statistics

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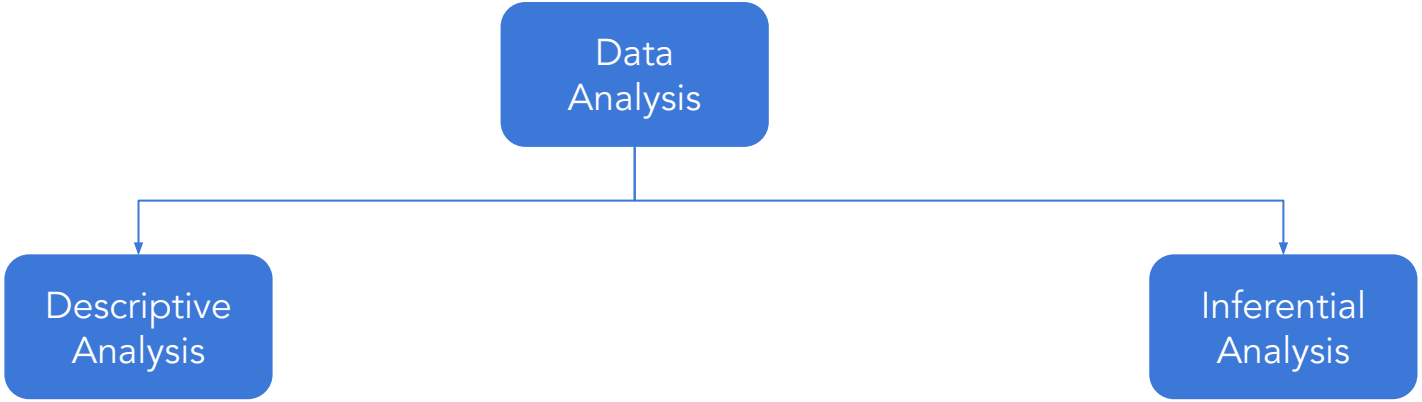
Data Analysis



- It is used to only describe the sample or summarize information about the sample
- Also known as Descriptive Statistics

- It is used to make inferences and generalizations about the broader population
- Also know an Inferential Statistics

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Population and Sample

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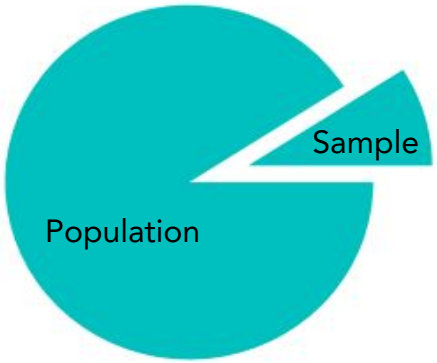
Population & Sample

- Population is a collection of all the individuals or objects

- Sample is a subset of the population which is a representative of the population



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Need to draw a sample

- In some of the real case scenarios we may not be able to gather all the observations in the population
- In such cases we can consider a sample which can be used to draw inferences about the population
- Sampling can reduce the computational time and cost
- In machine learning models, a sample obtained using an appropriate sampling strategy can provide accurate results with the less number of observations

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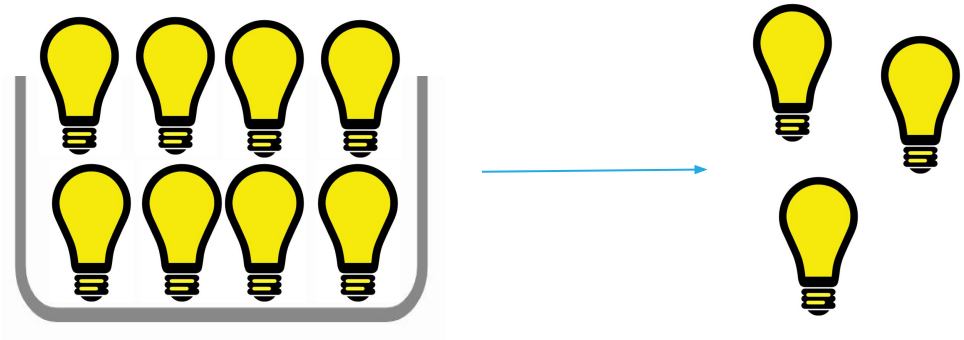
Need to draw a sample

Suppose a company producing electric bulbs wants to know the average life of a bulb. If the details of all the bulbs are available, then this information is regarded as the population.

It would be challenging for the company to test each and every bulb produced. In such a scenario, the company would draw a sample from the produced bulbs to test.



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Terminologies

- Sampling unit: An individual and indivisible unit of the population eligible to be in the sample
- Sampling frame: An exhaustive list of all members or elements of a population
- Sampling: To consider a subset of the population
- Sampling fraction (f): Ratio of sample size (n) to population size (N), i.e, $f = n/N$

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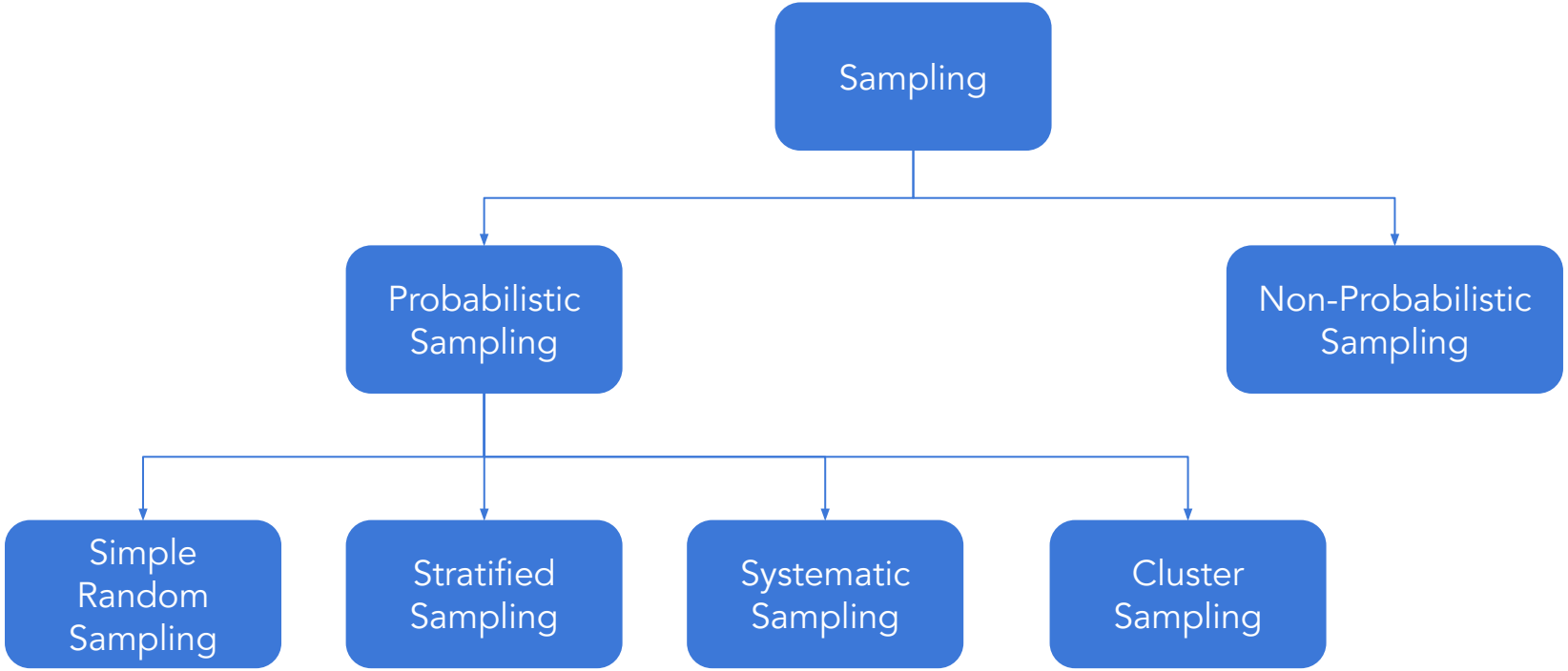
Sampling techniques



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Simple random sampling (SRS)

- Simplest and most common method
- Sample is drawn unit by unit
- Each unit has equal probability of being selected
- Two ways of selecting units:
  - Without replacement: if the selected unit is not returned to the population
  - With replacement: if the selected unit is returned to the population

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SRS without replacement

- At any specific stage in the sampling, the probability of any of the remaining units is 1/N, where N is the total number of sample units
- From N sample units, if a sample of size n is drawn, there are ${}^{N}C_{n}$ possible samples
- The probability of selecting one of them is $1/{}^{N}C_{n}$

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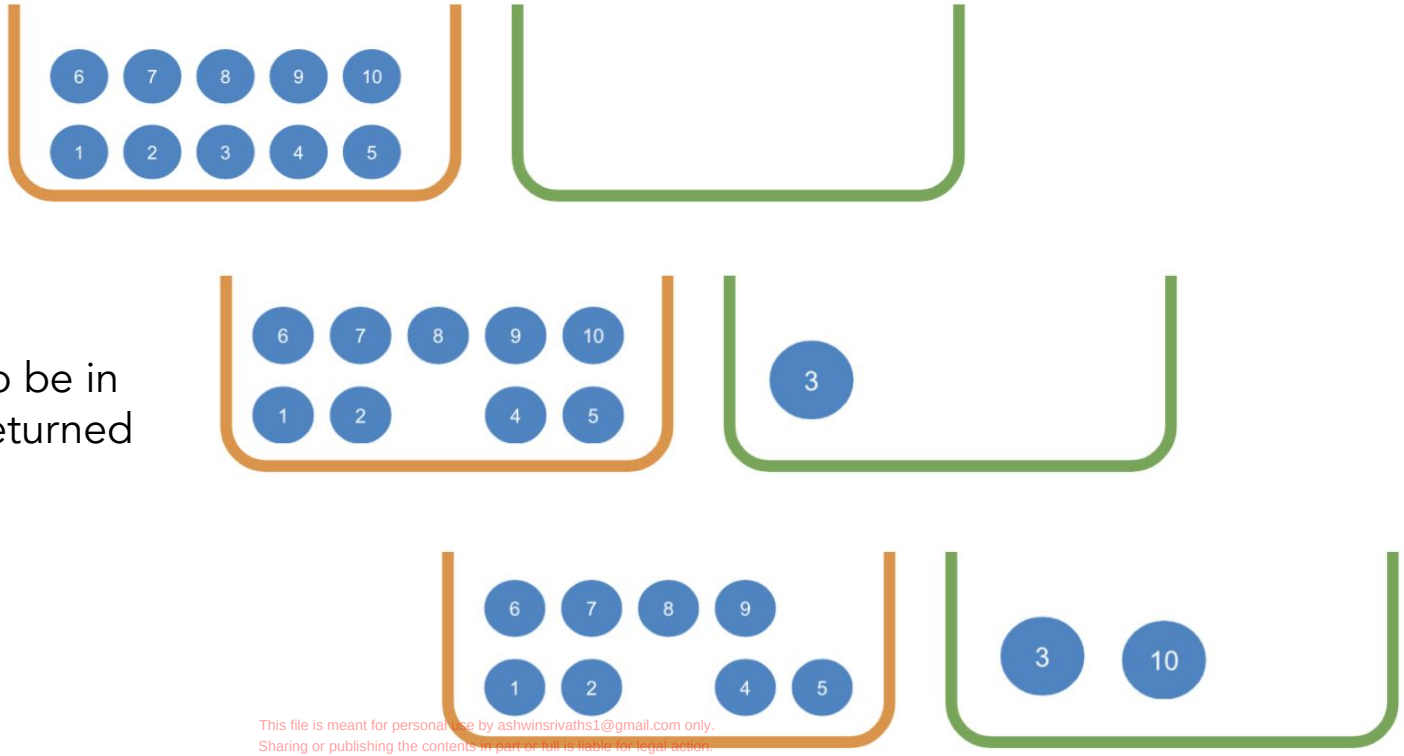
SRS without replacement

The unit selected to be in the sample is not returned to the population.



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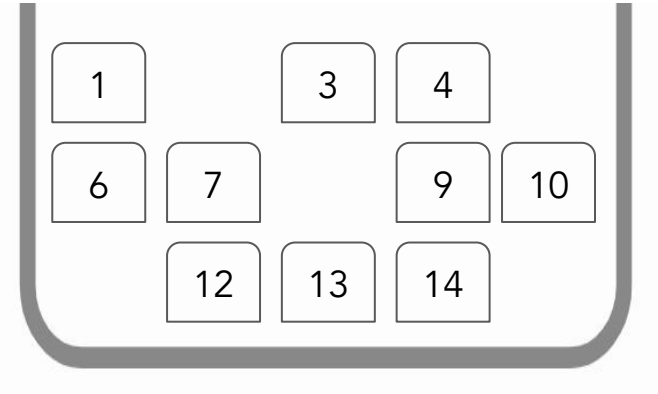
SRS without replacement - example

Drawing a sample from manufactured articles to know the proportion of defectives.

Manufactured articles



Sample to test for defective articles

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SRS with replacement

- The probability of selecting a specific unit at any given draw is always 1/N, where N is total number of sample units
- From N sample units, if sample of size n is drawn, there are N^n possible samples
- The probability of selecting one of them is $1/N^n$
- Example: You have 5 fruits and want to take a sample of 2. You can choose a fruit more than once.

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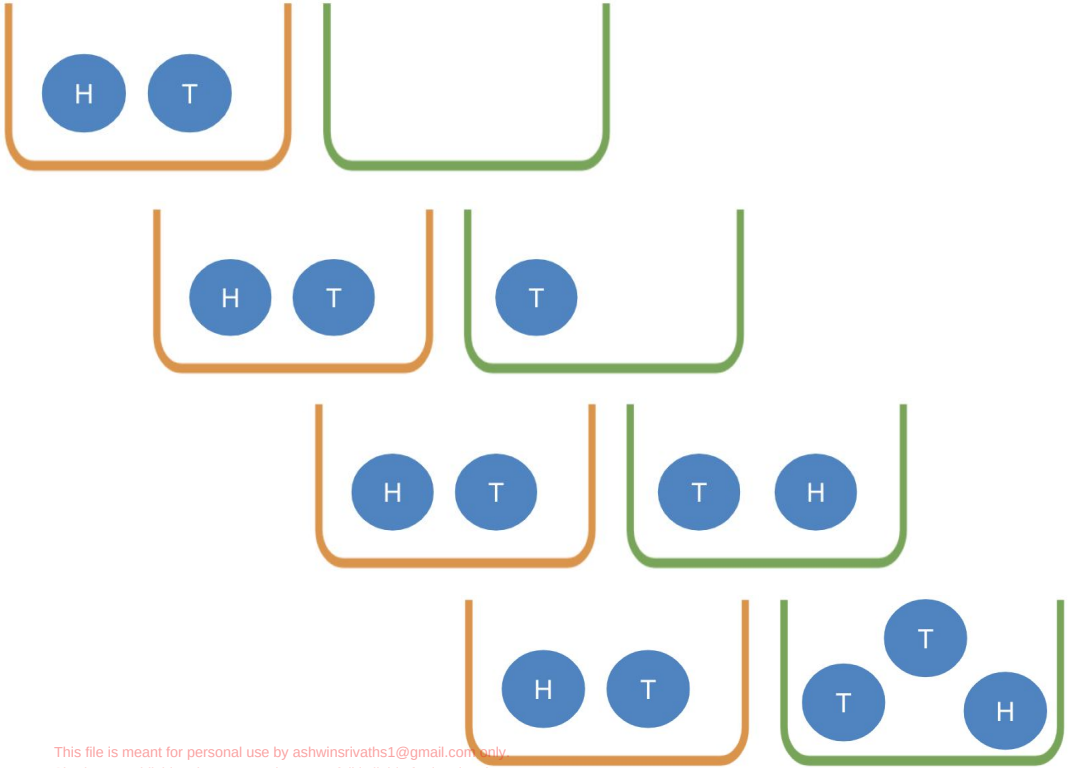
SRS with replacement

The unit selected to be in the sample is returned to the population.



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SRS with replacement - example

- Generally used in simulation studies

- A car insurance company wants to set the premium amount. The probabilities for major and minor accident are known to be 0.02% and 2% respectively
  (assume that an insuree can meet with an accident only once)

- A SRS with replacement is conducted to simulate 1 million such scenarios

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Simple random sample - python function

Python function	Description
random.choices()	draw sample with replacement
random.sample()	draw sample without replacement

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Sampling variability

- Let us draw four different samples from the data
- The sample mean varies from sample to sample
- This intuitive idea is called sampling variability
- The sample mean is close to the population mean

Population Mean
15.125

Sample	Sample Mean
14.7, 15.1, 15.0, 15.6, 14.9, 15.4, 15.8, 15.1	15.20
15.8, 15.5, 15.3, 15.4, 15.8, 14.8, 14.6, 14.8	15.250
15.7, 15.6, 14.8, 14.8, 14.7,14.9, 15.6, 15.1	15.150
14.8, 15.1, 15.7, 14.6, 15.6, 15.4, 15.6, 15.0	15.225

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PLEASE!! NOTE

A machine is used for packing grains in a sack. The weight of every bag is recorded as follows:

15.6	15.7	14.6	14.7	15.8	15.0	15.1	14.9
14.8	15.5	15.4	15.3	14.8	15.1	14.8	14.9

Draw at least the two samples and obtain their means. Compare this with the population mean.

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Summary

Sampling Method	Summary
Simple Random	Mostly used method. The sample can be selected with or without replacement.
Stratified	Sample point is selected from each subgroup in the population.
Systematic	Easy to implement for large population. The first sample point is selected randomly and remaining points are selected at fixed intervals.
Cluster	An entire subgroup in the population is selected as a sample point.

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Summary

- Sample is the subset of the population
- Different sampling methods: SRS with replacement (SRSWR), SRS without replacement (SRSWOR), Stratified, Systematic, Cluster sampling
- SRSWR method can be used if one wish to select the sample observation more than once
- Stratified sampling can be used when data is heterogeneous
- Cluster sampling can be used when the data consists of several groups

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Resampling

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Resampling

- Resampling is a technique in which we draw a sample from the available sample

- The resampling techniques are:
  - Bootstrap resampling
  - Jackknife resampling

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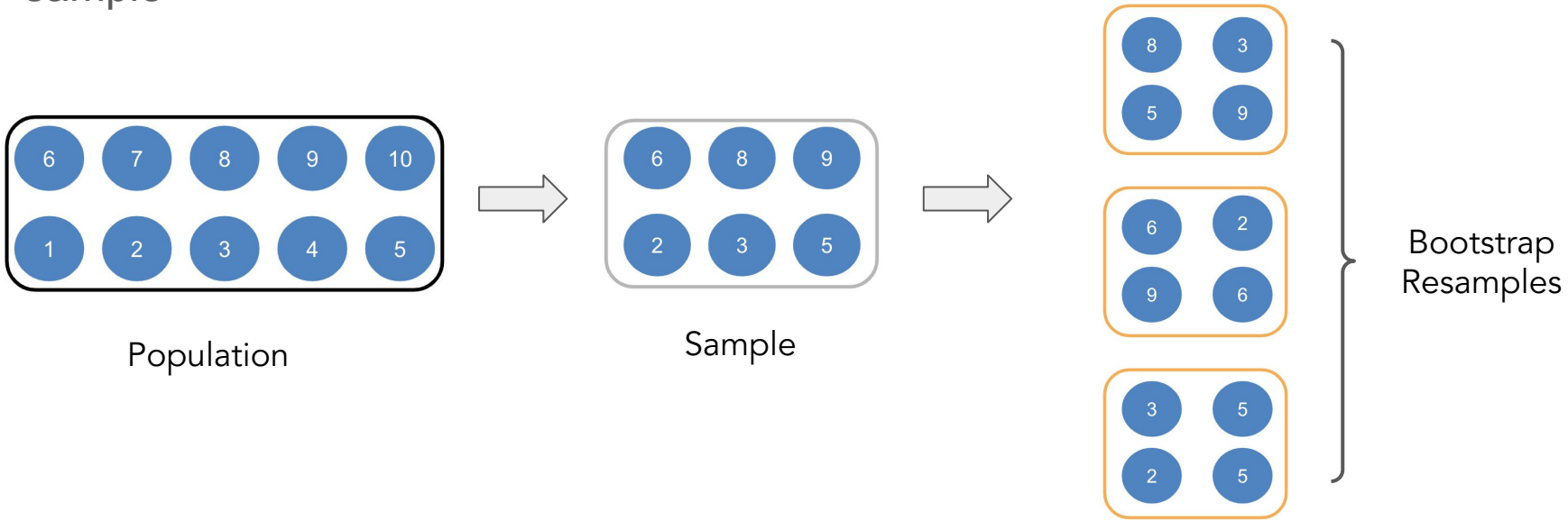
Bootstrap resample

Bootstrap resample is a simple random sample with replacement drawn from a sample



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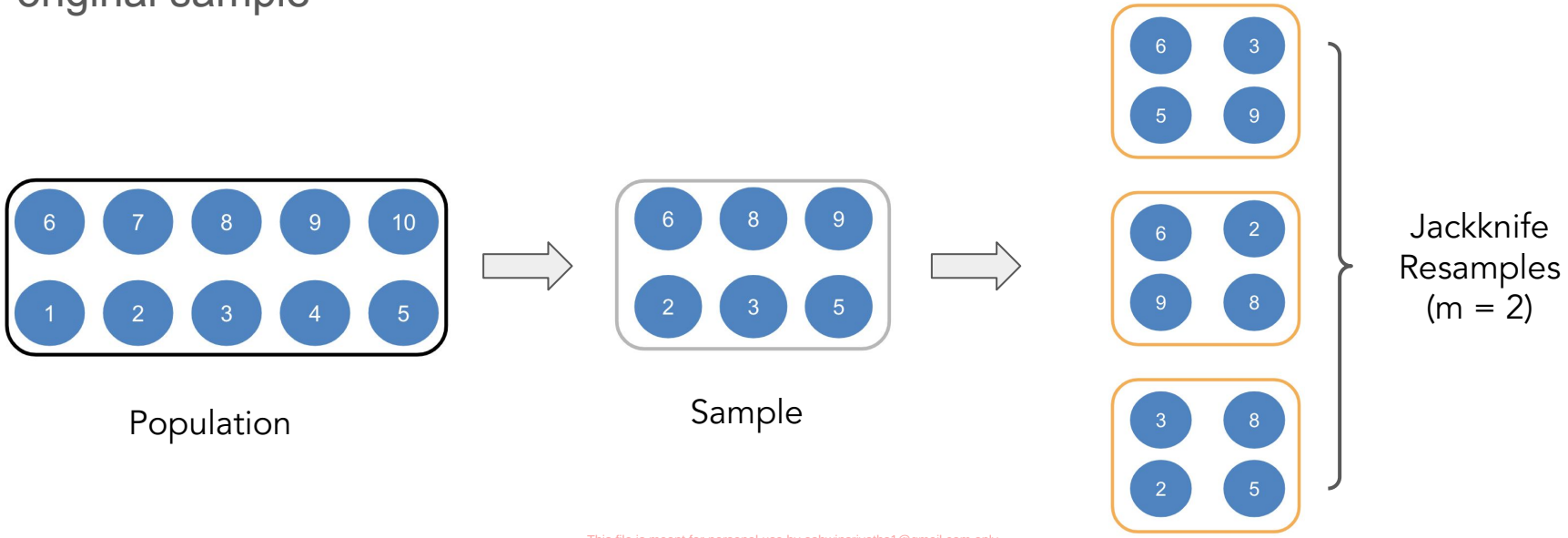
Jackknife resample

Jackknife resample is a method in which one deletes m observations from the original sample



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Descriptive Statistics

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What is descriptive statistics?

- Descriptive statistics is the term given to the analysis of the data that helps describe, visualize or summarize the data
- Helps exhibit the patterns in the data
- However, descriptive statistics only help in understanding the data and not make conclusions regarding any hypothesis
- We can not generalise the facts, they are confined only to the data at hand (that may be the sample)

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Descriptive statistics

The major characteristics we observe in each variable of the data set are:

1. Measures of central tendency
2. Distribution of the data
3. Measures of dispersion

The above characteristics help us in understanding the data and are part of the exploratory data analysis

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Measures of Central Tendency

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Measures of central tendency

- A measure of central tendency is a single value that identifies the central position of the data

- It includes mean, median, mode and partition values

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Mean

- Mean is a central tendency of the data
- It is defined as the sum of all sample observations divided by the number of observations in the sample:

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{N}$$

where N is the total number of observations

The whole data is spread out around the mean

Note: for sample data practices to use x (small x) and n (small n)

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Mean

Merits:

- Based on all observation

Demerits:

- Affected by extreme observations (outliers)
- Arithmetically it is not possible to obtain the mean of a data with missing values
- Mean can not be calculated for categorical data

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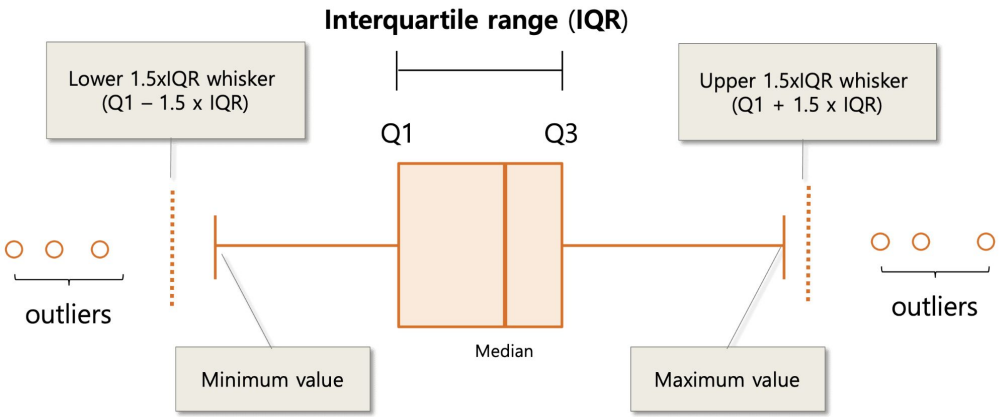
Outlier

- An outlier is a value that behaves differently than other observations
- It does not not follow the usual pattern



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Obtain the mean

Each column of a DataFrame is a pandas Series.

Let us create a series and find its mean.

```
prices = pd.Series([0,0,35,40,10,54,87,12,95,64,56,4,45,76,34,56,
    87,34,56,32,56,48,89,42,65,100,99,98,100,96,99])
```

```
# obtain the mean of 'prices'
prices.mean()
```

```
57.064516129032256
```

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Obtain the mean

Arithmetically, it is not possible to compute mean of missing data. But, python ignores the missing values and calculates the mean of the available data in the series

Introduce missing values in the previous series

```
# let us introduce some missing values in the series
# the remaining values are same as the series 'prices'
prices_missing = pd.Series([0,0,35,40,10,54,87,np.NaN,12,95,64,56,4,45,76,34,np.NaN,56,
                87,34,56,32,56,48,np.NaN,89,42,65,100,99,98,100,96,99])
```

```
# obtain the mean of prices_missing
prices_missing.mean()
```

```
57.064516129032256
```

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PLEASE!! NOTE

Mean of series with missing values

We notice that the means obtained are the same

This is because the function drops the missing values before calculating the mean value

```
# obtain the mean of 'prices'
prices.mean()
```

```
57.064516129032256
```

```
# obtain the mean of price_missing
prices_missing.mean()
```

```
57.064516129032256
```

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Trimmed mean

- Arithmetic mean obtained by ignoring lowest and highest α% observations is called as the α% - trimmed mean
- Using a trimmed mean, it is possible to eliminate the influence of outliers
- Note the observations are arranged either in increasing or decreasing order
- Also called as truncated mean

Example: The trimmed means are used in scoring the performance of an athlete during olympic games, to minimize the extreme scoring from possible biased judgements.

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Obtain the trimmed mean

While working in python, we can directly use the built in function, trim_mean(), available in the library scipy.

Import the required library

```
import scipy
from scipy import stats

# obtain a 20% trimmed mean
scipy.stats.trim_mean(prices, proportiontocut =0.20)
```

```
58.89473684210526
```

Parameter 'proportiontocut' denotes the α% values to exclude. It takes value between 0 to 0.51

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Median

- Median is the middle most observation in the data when it is arranged either in ascending or descending order of their values
- It divides the data into two equal parts. Thus, it is a positional average
- Median will be the middle term, if the number of observations is odd
- Median will be average of middle two terms, if number of observations is even

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Median

Merits:

- It is not affected by extreme values

Demerits:

- It is not based on all observations

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Obtain the median

```
prices.median()
```

```
56.0
```

Interpretation: The median value is 56. It implies that the value 56 divides the data in two equal parts. There are 50% of the observations above and below this price.

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Using mean and median

- For a numeric variable, the missing value is replace by the mean if there are no outliers present
- In case if outliers are present the missing values are replaced by the median
- For symmetric data median = mean, so either of the value can be used

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Mode

- Mode of the data is the value which has the highest frequency
- In other words, it is most repeated observation
- A set of observations having two modes is said to be bimodal
- A set of observations may have more than two modes. Such data is said to be multimodal

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Mode

Merits:

- It is applicable to both numeric and categorical data
- It is not affected by extreme observations

Demerits:

- It is not based on all observations

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Obtain the mode

```
prices.mode()
```

```
0    56
dtype: int64
```

0 is the index

Notice that the output is a Series

56 is the mode value

Interpretation: The mode value is 56. It implies that the value 56 occurs the most number of times

Also note that there is only one mode. The data is unimodal

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Obtain the mode

```
# create a series of pen prices
prices_pens = pd.Series([0,35,40,10,54,35,100,100])

# display the value counts of the
prices_pens.value_counts()
```

```
100    2
35     2
54     1
40     1
10     1
0      1
dtype: int64
```

Number of times each value is appearing in the data

Notice values 35 and 100 each occur 2 times

```
# print the mode
prices_pens.mode()
```

```
0     35
1    100
dtype: int64
```

The mode values

Interpretation: The mode values are 35 and 100. It implies that these values occur equal number of times and have the highest frequency.

Since there are two modes, the data is bimodal.

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Obtain the mode

```
# create a series of pen prices
prices_pens = pd.Series([0,0,35,40,10,54,76,34,56,87,34,56,100,99,100,99])

# display the value counts of the
prices_pens.value_counts()
```

```
56      2
99      2
100     2
34      2
0       2
76      1
10      1
40      1
87      1
54      1
35      1
dtype: int64
```

Number of times each value is appearing in the data

Notice values 0, 34, 56, 99, 100 each occur 2 times

```
# print the mode
prices_pens.mode()
```

```
0       0
1      34
2      56
3      87
4      99
5     100
dtype: int64
```

The mode values

Interpretation: The mode values are 0, 34, 56, 87, 99, 100. It implies that these values occur equal number of times and have the highest frequency.

Also note that there are 6 mode values. Data is multimodal.

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Partition values

- Partition values are the values which divide the dataset into equal parts
- Note that partition values may not be equispaced
- Also known as quantiles
- We have seen that the median divides the data into two equal halves. Similarly in order to make four parts we use quartiles, for ten parts we use deciles

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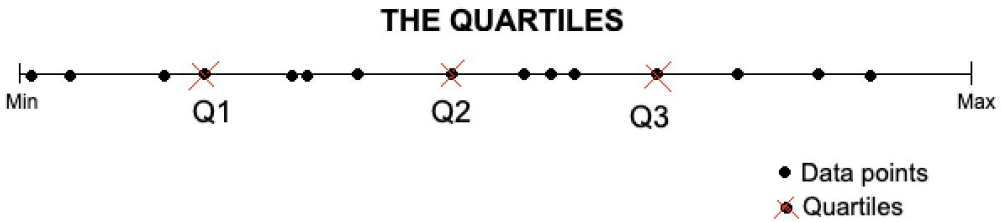
Quartiles

- The values that divide the dataset into four equal parts are called quartiles



- 25% of the data lies below the first quartile (Q1) and 75% above it
- The second quartile divides the data points in two equal halves. Q2 is the median
- The third quartile (Q3) divides the the observation in 75%-25% meaning that 75% of the data lies below Q3 and 25% above it

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Obtaining the quartiles

The partition values are obtained using the quantile function specifying the percentage of data below the required partition value.

```
# to get the first quartile
prices.quantile(.25)
```

```
34.5
```

```
# to get the second quartile
prices.quantile(.50)
```

```
56.0
```

```
# to get the third quartile
prices.quantile(.75)
```

```
88.0
```

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Deciles

- The values that divide the dataset into ten equal parts are called deciles
- 10% of the data lies below the first decile (D1) and 90% above it
- Similarly, 20% of the data lies below the second decile (D2) and 80% above it and so on
- The fifth decile is the same as the second quartile i.e. D5 = Q2 = Median
- There are in all 9 deciles

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Obtain the nine deciles (D9) for the variable ‘prices’ defined before.

Similarly we define percentiles as values which divides the data into 100 equal parts. Obtain the 25th (P25), 50th (P50) and 75th (P75) percentile.

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Summary

- Mean, median and mode can be used to impute the missing values in the data
- For a symmetric distribution, mean = median = mode
- For any data,

$$Q2 = D5 = Median = P50$$

- Mean is affected by the presence of outliers
- Trimmed mean is useful when the outliers are present in the data
- Median and mode are not affected by the presence of outliers

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Distribution of the Data

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Distribution of the data

- The distribution is a summary of the frequency of values taken by a variable
- The distribution of the data gives information on the shape and spread of the data
- On plotting the histogram or a frequency curve for a variable, we actually look at how the data is distributed over its range

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Distribution of the data

- The data may be distributed in a random matter
- However there are standard distributions in statistics like the normal distribution, binomial distribution, student’s-t distribution and so on

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Measures of Dispersion

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Measures of Dispersion

- The measure of dispersion refers to how the data is spread around the central value for a given variable.

- Reliability of a central value inversely depends on the degree of spread of the data.
  - Larger the spread, less reliable the central value

- The spread of data is measured using range, variance, standard deviation, coefficient of variation and interquartile range.

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Range

- Range is the difference between the largest and the smallest observation
- It is defined as

$$\text{Range} = X_n - X_1$$

where X_n is the largest value and X_1 is the smallest value

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Range

Merits:

- It is a basic way to measure the variability

Demerits:

- If either the smallest or the largest value turns out to be an outlier then the range value is unreliable
- The other in-between observations do not affect the range value
- It does not give a clear picture of variability in the data

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Obtain the range

```
# Series 1
year_establishment = pd.Series((1955, 1900, 1980, 1990, 2000))
Range = year_establishment.max() - year_establishment.min()
Range
```

```
100
```

```
# Series 2
year_establishment = pd.Series((1920, 1990, 1925, 1930, 2020))
Range = year_establishment.max() - year_establishment.min()
Range
```

```
100
```

Use the formula for range

Interpretation: The range of both the variables is 100. However, it seen that the first series has observations towards the maximum and the second series has observations towards the minimum.

Thus, range is not a good measure of dispersion.

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Variance

Variance in a sample data is the arithmetic mean of squared deviations of the data from the mean. It is given as

Population Variance	Sample Variance
$\sigma^2 = \dfrac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$	$s^2 = \dfrac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$
σ^2 = population variance	s^2 = sample variance
x_i = value of ith element	x_i = value of ith element
μ = population mean	$\bar{x}$ = sample mean
N = population size	n = sample size

Interpretation: The higher the variance more the data is spread out.

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Obtain the variance

```
# obtain the variance for the variable 'prices'
prices.var()
```

```
1054.3290322580644
```

More the data is spread out away from the mean, higher the variance.

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The unit of measurement of variance

- The unit of measurement of the variance is not the same as that of the original data
- Consider an industry producing screws. The diameter of each screw is noted in millimeters
- The variance in the diameter is measured in squared millimeters
- Thus a usual practice is to take square root of the variance which would have the same unit as the original data
- The value thus obtained is called the standard deviation of the data

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Standard deviation

The standard deviation of the variable is the square root of the variance.

$$\text{Standard deviation } (\sigma) = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \bar{x})^2}{N}}$$

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Obtain the standard deviation

```
prices.std()
```

```
32.470433200960905
```

Interpretation: The standard deviation looks at how spread out the observations are from the mean. The higher the standard deviation more the data is spread out.

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Coefficient of variation

- The coefficient of variation is a statistical measure of dispersion of data points around the mean

$$\text{Coefficient of Variation (C.V.)} = \frac{Standard\ Deviation}{|Mean|} X\ 100 = \frac{\sigma}{|\bar{x}|} X\ 100$$

- It is always expressed in percentage and is unit free
- Generally, used to compare dispersion of two or more groups
- According to Karl Pearson, C.V. is the percentage variation in the mean and S.D. is the total variation in mean

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Obtain the coefficient of variation

```
# importing variation from scipy.stats
from scipy.stats import variation

scipy.stats.variation(prices)
```

```
0.5597598237090757
```

The variation() from the scipy library computes the coefficient of variation.

Note: The name of the function can be quite misleading; do not confuse with 'variance' and coefficient of correlation

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Obtain the coefficient of variation

Consider two financial portfolios, A and B. Obtain the the better portfolio in which an investor should invest.

Portfolio	Expected Return (%)	Volatility (%)
A	25	30
B	10	15

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Obtain the coefficient of variation

Obtain the coefficient of variation for the two portfolios

```
# the expected return of portfolio A (in %)
expected_return_A = 25

# the volatility of portfolio A (in %)
volatility_A = 10

# obtain the coefficient of variation
cv =  volatility_A/expected_return_A

# print the coefficient of variation
cv
```

```
0.4
```

```
# the expected return of portfolio B (in %)
expected_return_B = 30

# the volatility of portfolio B (in %)
volatility_B = 15

# obtain the coefficient of variation
cv =  volatility_B/expected_return_B

# print the coefficient of variation
cv
```

```
0.5
```

Interpretation: The coefficient of variation for portfolio A is lower than that of portfolio B.

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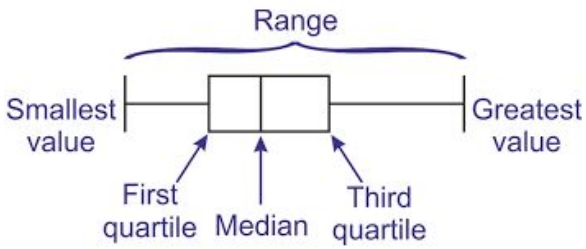
Quartiles

- We have seen quartiles are kind of partition values
- With the help of the quartiles we understand the spread of the data
- Quartiles when plotted with box plot help in identifying the outliers



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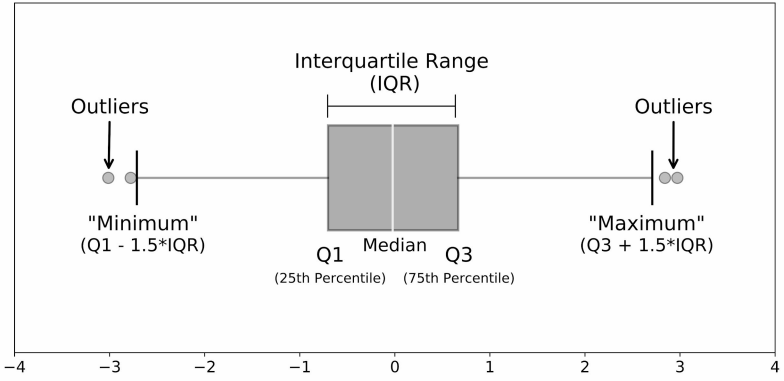
https://rahulsblogorg.files.wordpress.com/2017/08/whisker.png

Understanding box plot

- A box plot is a way of displaying the distribution of data based on a five number summary (minimum, first quartile (Q1), median, third quartile (Q3), and maximum)

- The whiskers give an idea of the spread of the data

- The dots outside of the whiskers are the outliers



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Interquartile range

- The interquartile range is defined as the difference between the third and the the first quartile
- It gives the range in which the middle 50% of the data lies
- The value of IQR is used to identify outliers in the data

$$\mathrm{IQR} = \mathrm{Q}_3 - \mathrm{Q}_1$$

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Obtain interquartile range

```
# obtain the first quartile
Q1 = prices.quantile(0.25)

# obtain the quartile
Q3 = prices.quantile(0.75)

# obtain the IQR
IQR = Q3 - Q1

# print IOR
IQR
```

```
53.5
```

Interpretation: The interquartile range of the prices is 53.5

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Z-score

- Every value in the dataset has the z-score
- To find the z-score of a value, subtract the mean from it and divide it by the standard deviation
- The z-scores are extensively used in statistics especially in hypothesis testing
- It is also used to normalize the data, ie is it scales the data to get the mean of the scaled data as 0 and variance as 1

$$\text{Z- score} = \frac{x_i - \bar{x}}{\sigma}$$

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Z-score

The z-score tells how much is the value away from the mean

Z-score Value	What it means?
0	The corresponding value is same as the mean
< 0	The corresponding value is less than the mean
> 0	The corresponding value is greater than the mean
< -3 or > 3	The corresponding value is potential outlier

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Summary

- Range is not a reliable measure as it do not consider the in-between observations
- The variables with a near-zero standard deviation have less significance in the analysis
- Coefficient of variation (CV) is the unitless quantity
- Box-plot is used to visualize five-number summary (minimum, Q1, Q2, Q3, and maximum)
- IQR can be used to detect and remove the outliers in the data

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Skewness

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Skewness

- Skewness is a lack of symmetry or departure from symmetry
- If the distribution of the data is elongated in any one of the sides then the data is said to be skewed
- If the distribution of the data is elongated on the left side (negatively skewed) then the data is said to be left skewed
- If the distribution of the data is elongated on the right side (positively skewed) then the data is said to be right skewed

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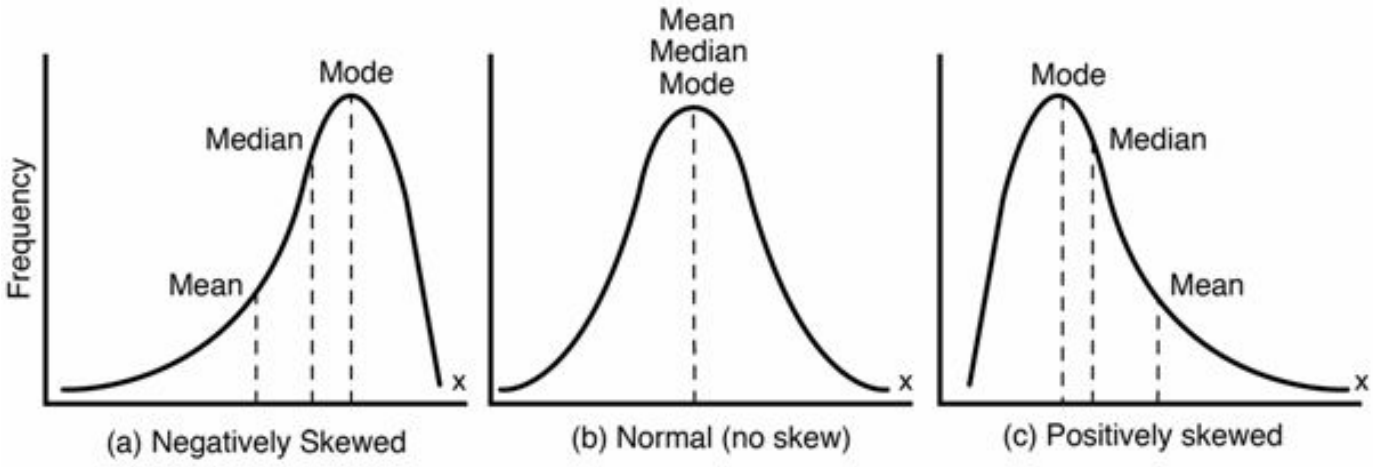
Skewness

The graph of skewed distribution



mean < median < mode mean = median = mode mean > median > mode

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Calculate the skewness

The Bowley's Coefficient of Skewness is defined as

$$S_b = \frac{Q_3 - 2\,Q_2 + Q_1}{Q_3 - Q_1}$$

where Q_1, Q_2 and Q_3 are first, second and third quartiles respectively

Value of skewness	Interpretation
$S_b < 0$	The distribution is negatively skewed
$S_b = 0$	The distribution is not skewed (symmetric distribution)
$S_b > 0$	The distribution is positively skewed

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Calculate the skewness

The Bowley's coefficient of skewness:

```
# calculate Bowley's coefficient of skewness
# calculate first quartile
Q1 = prices.quantile(0.25)

# calculate second quartile
Q2 = prices.quantile(0.5)

# calculate third quartile
Q3 = prices.quantile(0.75)

s_b = (Q3 - 2*Q2 + Q1) / (Q3 - Q1)

# Bowley's coefficient of skewness
print(s_b)
```

```
0.19626168224299065
```

Interpretation: The coefficient is positive but near zero. Thus, the variable 'prices' is said to be near symmetric.

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Calculate the skewness

The Karl Pearson's Coefficient of Skewness is defined as

$$S_k = \frac{3(mean - median)}{\sigma}$$

Where σ is the standard deviation

Value of skewness	Interpretation
$S_k < 0$	The distribution is negatively skewed
$S_k = 0$	The distribution is not skewed (symmetric distribution)
$S_k > 0$	The distribution is positively skewed

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Calculate the skewness

The Karl Pearson's coefficient of skewness:

```
# calculate Karl Pearson's coefficient of skewness
# calculate the mean
price_mean = prices.mean()

# calculate median
price_med = prices.median()

# calculate standard deviation
price_sd = prices.std()

s_k = (3*(price_mean - price_med)) / price_sd

# Karl Pearson's coefficient of skewness
print(s_k)
```

```
0.09835250325524642
```

Interpretation: The coefficient is positive but near zero. Thus, the variable 'prices' is said to be near symmetric.

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Obtain the skewness

In python, the ‘skew()’ returns the value of skewness in the distribution

```
# use the 'skew()' to calculate the skewness
prices.skew()
```

```
-0.20497188990720197
```

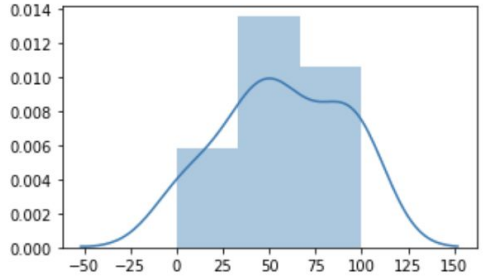
Interpretation: The variable prices is slightly negative skewed. It can be said it is close to symmetric.

```
# import the required libraries
import seaborn as sns
import matplotlib.pyplot as plt

# set the plot size
plt.figure(figsize=(5,3))

# plot a distribution plot
sns.distplot(prices)

# display the plot
plt.show()
```



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PLEASE!! NOTE

Coefficient of skewness

- The magnitude of the coefficient of skewness explains the extent of skewness in the distribution
- The +/- sign of a coefficient provides the direction of skewness
- The coefficient compares the distribution of the sample to the normal distribution. Thus, the zero value represents the symmetric distribution
- A higher magnitude of the coefficient indicates that the distribution highly differs from the normal distribution

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Summary

Bowley's Coefficient of Skewness

- Based on quartiles
- Value lies between -1 to +1
- Can be used to calculate skewness for open-ended distribution
- Preferred if the data has extreme values (outliers)

Karl Pearson's Coefficient of Skewness

- Based on mean, median and standard deviation
- Usually value lies between -3 to +3
- Can not be used for open-ended distribution, as we can not calculate the mean in such case

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Kurtosis

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Kurtosis

- Kurtosis measures the peakedness of the distribution
- In other words, it is a statistical measure that defines how the tails of the distribution differ from the normal distribution
- Kurtosis identifies whether the tails of a given distribution contain extreme values
- A histogram is used as a estimation of density is an effective way to show both the skewness and kurtosis of a data set

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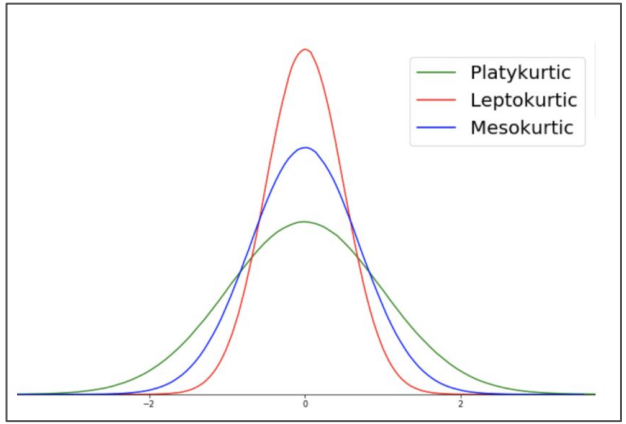
Kurtosis

Value	Thickness of Tails	Interpretation
Kurtosis < 0	Thin	The distribution is platykurtic
Kurtosis = 0	Normal	The distribution is mesokurtic
Kurtosis > 0	Thick	The distribution is leptokurtic



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Obtain the kurtosis

We use kurt() to find the kurtosis

Interpretation: The variable ‘prices’ is platykurtic since -1.036 < 0. This means there are more dispersed prices with lighter tails. i.e. less extreme values.

```
prices.kurt()
```

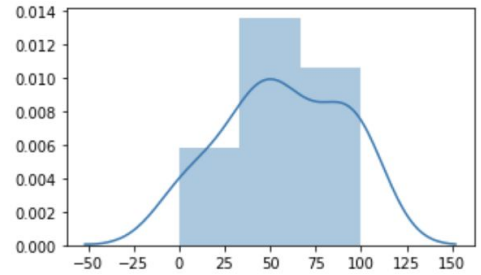
```
-1.0366224980536156
```

```
# import the required libraries
import seaborn as sns
import matplotlib.pyplot as plt

# set the plot size
plt.figure(figsize=(5,3))

# plot a distribution plot
sns.distplot(prices)

# display the plot
plt.show()
```



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Summary

- A mesokurtic distribution is basically a normal distribution
- A leptokurtic distribution has heavy tails on both the sides that indicates the presence of large outliers. Also the distribution is concentrated towards the mean than the normal distribution
- A platykurtic distribution has flat/lighter tails on both the sides and it indicates that there are small amount of outliers in the distributions. Also, the values are more dispersed and fewer values are close to the mean

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Covariance

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Covariance

- Covariance is a measure of how much two random variables vary together
- Covariance is similar to variance, but variance explains how a single variable varies, and covariance explains how two variables vary together
- It cannot tell whether the value indicates a strong relationship or weak relationship, since the covariance can take any value

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Covariance

The covariance for two variables X and Y in sample data is given by

$$COV(X,Y) = \frac{\sum_{i=1}^{n}\left(X_i - \overline{X}\right)\left(Y_i - \overline{Y}\right)}{n-1}$$

Xi = values taken by variable X , $\forall$ X $\in$ [1, n]

Yi = values taken by variable Y , $\forall$ Y $\in$ [1, n]

$\overline{X}$ = mean of Xi

$\overline{Y}$ = mean of Yi

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Obtain the covariance

```
# create imports series
imports_raw_material = pd.Series([10,11,14,14,20,22,16,12,15,13])

# create exports series
exports_finished_products = pd.Series([12,14,15,16,21,26,21,15,16,14])

# calculate the covarinace
imports_raw_material.cov(exports_finished_products)
```

```
15.444444444444443
```

Interpretation: There is a positive covariance between imports of raw material and export of finished goods

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Correlation

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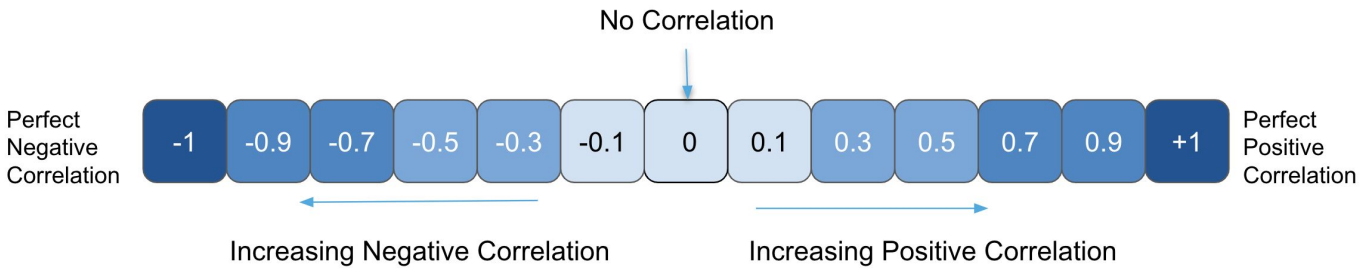
Correlation

- Correlation is the standardized version of covariance (covariance/respective correlation).
- Correlation is the extent of linear relationship among numeric variables
- It indicates the extent to which two variables increase or decrease in parallel
- The value of a correlation coefficient ranges between -1 and 1



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Correlation

- Correlation is normalized form of covariance

$$Corr(X,Y) = \frac{Cov(X,Y)}{\sigma_x \sigma_y}$$

where σ_x is the standard deviation of X and σ_y is the standard deviation of Y

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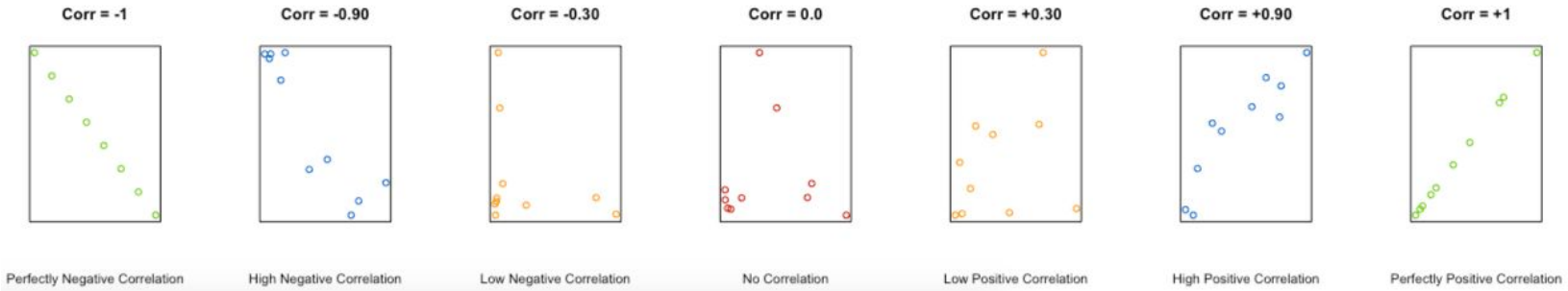
Correlation

- A positive correlation exists if variables increase/decrease together
- A negative correlation exists if one variable increases, while the other variable decreases



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Obtain the correlation

```
# create imports series
imports_raw_material = pd.Series([10,11,14,14,20,22,16,12,15,13])

# create exports series
exports_finished_products = pd.Series([12,14,15,16,21,26,21,15,16,14])

# calculate the correlation
imports_raw_material.corr(exports_finished_products)
```

```
0.9458496510416226
```

Interpretation: There is a high positive correlation between imports of raw material and export of finished goods

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Covariance and correlation

Covariance

- tells whether two variables are related by measuring how the variables change in relation to each other
- Gives the direction of the relationship

Correlation

- Similar to covariance it gives the direction of the relation
- It also tells the strength is the relationship

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Measures of central tendency

A measure of central tendency is a single value that identifies the central position of the data

Measure of Central Tendency	Description	Example
Mean	Ratio of sum of all observations to the number of observations	price = [1, 2, 3, 4, 5] mean = 3
Median	Middle most observation in the data when it is arranged either in ascending or descending order of their values	price = [6, 8, 9, 18, 20] median = 9
Mode	Value which has the highest frequency	price = [6, 18, 9, 18, 20] mode = 18

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Measures of central tendency

Measure of Central Tendency	Description
Partition values	The values that divide the data into desired number of equal parts
Quartiles	The values that divide the data into four equal parts

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Measures of dispersion

The measure of dispersion refers to the variability within the data where variability is the measure of how close or far the data lie from the central value

Measures of Dispersion	Description	Example
Range	The difference between the largest and smallest value in the data	price = [1, 2, 3, 4, 5] range = 4
Variance	The arithmetic mean of squares of deviations taken from the mean	price = [6, 18, 9, 18, 20] variance = 31.36
Standard deviation	The square root of variance	price = [6, 18, 9, 18, 20] Std deviation = 5.6

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Measures of dispersion

Measures of Dispersion	Description
Coefficient of variation	A relative measure of deviation, it is the ratio of standard deviation to the mean
Interquartile Range	The difference between the first and the third quartile

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Skewness and kurtosis

	Description
Skewness	Lack of symmetry or departure from symmetry
Kurtosis	Measures the peakedness of the distribution. In other words, defines how the tails of the distribution differ from the normal distribution

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Covariance and correlation

	Description
Covariance	Tells whether two variable are related or not and gives the direction of the relationship (positive/negative)
Correlation	Tells whether two variable are related and along with the direction of the relationship (positive/negative), it tells how strong the relationship is

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